



মিউচুয়াল ট্রাস্ট ব্যাংক পিএলসি
Mutual Trust Bank PLC
you can bank on us

Remittance API

Partner Integration Guide

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Audience: MTO/Money Exchange House/Payment Facilitator/Payment Aggregator

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Introduction

1.1 Preface

The MTB Remittance API is a customized implementation in context of the local & International Remittance market's demands and business operation needs. The Platform can leverage integrations with various trusted FI (Financial Institution) like Bank, Money Transfer Organizations, Money Exchange Houses, Payment Facilitator, Payment Aggregator and MFS through the defined interfaces to implement services like borderless Real Time Transaction, so that a FI registered users can send real time inward remittance to any Bangladeshi Bank/Wallet account.

1.2 Purpose of Document

The purpose of this document is to enable partners to directly integrate with MTB systems to develop remittance services as per defined scope of engagement.

This document describes the interface for trusted FI systems covering communication protocols, security requirements, call flows and functions to perform transactions, actions, business operations and queries.

1.3 Audience

The intended audiences of this document are:

- Solution Architect
- System Engineer
- Application Developer
- Integration Tester

Preconditions

2.1 Environment Setup

2.1.1 Staging (UAT)

For acceptance testing, all access to APIs will be provided on the MTB Remittance Solution UAT system. Test partner accounts will be setup for API implementation and testing. Partners must successfully pass Technical and Business UAT on the UAT setup before proceeding to production.

2.1.2 Production (Live)

Once all API validations are through and UAT is successful, partners will be on-boarded on the production setup. Production account details will be shared with the partner for Go-Live.

2.1.3 API URL

Live: api.mutualtrustbank.com/remit

UAT: uat.mutualtrustbank.com/remitapi

2.2 Connectivity Establishment

MTB Remittance Solution API is hosted at MTB on premise. Partners must establish connectivity with the MTB server which is publicly exposed. No whitelisting is required for UAT.

All API interfaces between the MTB Remittance Solution API and integrating partner will be over HTTPS.

Steps involved:

- MTB will share domain name to partners for both UAT and LIVE
- Partner shall share IP and other necessary information in the circulated form (For LIVE)
- MTB to whitelist Partner's public IP/s for LIVE connectivity.

Integration Overview

3.1 API Overview

API enables MTB Remittance Solution system to integrate with Partner system.

- All APIs are invoked based on the Restful protocol over HTTPs.
- Message format: JSON
- The entire API supports synchronous mode of API call.
- Error Codes will be shared which are to be mapped into Partner's system.
- Error codes are subject to be added from time to time.

The API defines:

- The message transfer sequence between a Partner and MTB system.
- A set of API interfaces are available for initiating business operations or transactions between a FI and MTB system.

3.2 API List

API Name	Method	Details
Access Token	POST	Generates token by User Id, Party Id and Password.
Exchange House Details	POST	Get Exchange house NRT Account(s), Active Status, Transaction Mode availability, Available Balance
Change Password	POST	Exchange house can change their password
Account Validation	POST	Perform MTB & Wallet Account Validation. Not Mandatory for Actual Payment
Bill Information Validation	POST	Get Utility Bill Information. Mandatory for Actual Payment
Payment Request	POST	Receive Transaction requests for all type of transactions. Tran Mode can be differentiated by dedicated parameter(s)
Branch Information	POST	Branch Name, Branch Code, Routing Number, Branch List etc for BEFTN/RTGS/NPSB
Query Payment Status	POST	Fetch the Transaction summary based on the parameter (Date, reference , state, payment type, etc)
Account Statement	POST	Account Statement in Core Banking System with CBS Date and Accounting Type

3.3 Conversation Management

- A Request Id is to be generated by the Partner to identify a unique conversation inside the MTB Remittance system. The Request Id at the MTB Remittance Solution side is used to relate the request & response for a single interaction. It cannot be duplicated for the whole life cycle of the system.
- Partner can query for troubleshooting or other purposes just by providing the Request Id.
- MTB Remittance System is a 2 tier architectural system where first layer is this API system. MTB receives, validates and then saves the payment data through API request. MTB Internal Remittance Management System then processes those payment requests, performs necessary accounting and forwards the same to respective System (OtherBank/Wallet/Customer).
- As such, there are two types of codes to be mapped in Partner System.
 - **API Response Code** --- Response from each of APIs
 - **Status Code** --- Actual Payment Status (Derived from StatusQuery API)
- Response Code and Status Code will be available in a separate Document.

Authentication & Security

Data Security measure should be adopted at both end during in use & store the API request credential & sensitive information.

4.1 Digital Certificate Authentication

Item	Description
SSL Authentication	- Supported SSL versions: TLS 1.2 & Onwards. - One way Authentication - Public Key Cryptography Standards (PKCS): PKCS#12

4.2 JWT Authentication

- Every API request requires server side jwt Token Validation. The token has to be passed as *Bearer Token* in Request authorization.
- Token will be generated through **AccessToken** API using UserId and Password (Pre-shared by MTB).
- Partner can change the password anytime using **ChangePassword** API
- Each token expires after 60 minutes.

4.3 Sensitive Data Encryption

- As part of message layer security, some sensitive parameter value will be encrypted in the Request & Response.
- Symmetric key based encryption will be used to encrypt & decrypt parameters value. Different Partner will have different group of Keys.
- MTB will provide Request Key and Response Key separately.
- Partner (ExchangeHouse) will encrypt sensitive Request Data with 'Request Key' and MTB will decrypt those data using the same key. While sending Response, MTB will encrypt sensitive Response Data with 'Response Key' and Partner will decrypt those data using the same key.
- Sensitive parameter definition has been given in the API specifications.
- 256 bit AES will be used to encrypt the sensitive parameters.

- Cipher block chaining (CBC) technique will be used for cipher text.
- Dynamic IV will be used in the encryption methodology with following arrangement:
- AES Key: **X** bit IV (Random) + **X** bit Key offset (fixed). Key offsets will be pre-shared.
- For encryption, Sender will generate **X** bit IV & form the AES key (IV + Key Offset). Then sender performs the encryption of sensitive parameters value using the AES KEY.
- After encryption, sender add the generated IV with encrypted value as prefix (left padding) & send the information to receiver as parameter value in base64 encoded format.
- Upon receiving, receiver will separate the **X** bit IV from the received parameter value.
- Now receiver can form the AES KEY to decrypt the actual encrypted value.
- MTB will share Request Key, Response Key and Key Offset (Value of **X**).

API Specification

5.1 Access Token

Controller: token

End Point: {URL}:<PORT>/<AliasName>/<ControllerName>/accessToken

Authorization:

Basic Authorization ---

username: <will_be_provided>

password: <will_be_provided>

Method: POST

Request Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
remitChannelId	string	M	A default value	N	MTB will share this value. Will be unique for whole life cycle for each Partner.

Example of Request Body

```
{
  "remitChannelId": "Q3h0qECKr6QwG2h2tJ+avMv23ZqChc="
}
```

Response Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
respCode	string	M		N	RS0000 for success
respDesc	string	M		N	
responseDetails	object				
accessToken	string	O	token	N	JWT Token
tokenType	string	O		N	
expiryAfter	string	O	Will expire after 60 minutes	N	Field contains value in seconds

Example of Response Body

```
{
  "respCode": "RS0000",
  "respDesc": "Request Successful",
  "responseDetails": {
    "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdXkiOiJMSlMvV4cCI6MTY1NjIzNTgyMCwiaXNzIjoiaHR0cDo"
  }
}
```

```
vL2xvY2FsaG9zdDo0NTM5NCIsImF1ZCI6Imh0dHA6Ly9sb2NhbgGhvc3Q6NDUzOTQifQ.JTEAgCDgiPKx4gJMw36t
YQmpL0XUB3szZQH1-0aT-SY",
  "tokenType": "Bearer",
  "expiryAfter": "3600"
}
```

5.2 Exchange House Details

Controller: extension

End Point: {URL}:<PORT>/<AliasName>/<ControllerName>/partnerDetails

Authorization: Bearer Token

Method: POST

Request Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M	Request ID. Alphanumeric in format with no special characters. Indicates the unique identifier of the request message generated by Partner.	N	This Id will be used to handling duplicate & end to end message tracing & trouble-shooting. (Every API must have this id and it must be unique for whole life cycle)

Example of Request Body

```
{
  "reqId": "MTB1234567890Aqsrt"
}
```

Response Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M		N	Request ID received in the corresponding request Body
respCode	string	M		N	RS0000 for success
respDesc	string	M		N	
responseDetails	object				
nrtAccount	string	O	13 Digit MTB account	Y	
availableBalance	string	O		Y	Balance in BDT
tranModeActive	object				
wageEarningsRemittance	string	O	Partner is permitted to send WageEarner Remittance	N	true/false
serviceRemittance	string	O	Partner is permitted to send Service Remittance	N	true/false
b2bRemittance	string	O	Partner is permitted to send Business to Business Remittance	N	true/false

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utilityRemittance	string	O	Partner is permitted to send Business to Business Remittance		true/false
otherBankAccount	object				
beftn	string	O	Partner is permitted to send BEFTN	N	true/false
rtgs	string	O	Partner is permitted to send RTGS	N	true/false
npsb	string	O	Partner is permitted to send NPSB	N	true/false
mtbAccount	string	O	Partner is permitted to send OwnBank Transfer	N	true/false
cash	string	O	Partner is permitted to send Cash Transaction	N	true/false
wallet	object				
bKash	string	O	Partner is permitted to send bKash Wallet Transaction	N	true/false
Nagad	string	O	Partner is permitted to send Nagad Wallet Transaction	N	true/false
Upay	string	O	Partner is permitted to send Upay Wallet Transaction	N	true/false
iPay	string	O	Partner is permitted to send iPay Wallet Transaction	N	true/false

Example of Response Body

```
{
  "reqId":"MTB1234567890Aqsrt ",
  "respCode":"RS0000",
  "respDesc":"Request Successful",
  "responseDetails":{
    "nrtAccount":"0460310012345", // ----- will be encrypted
    "availableBalance":"1000.00", //----- will be encrypted
    "tranModeActive":{
      "wageEarnersRemittance":"true",
      "serviceRemittance":"true",
      "b2bRemittance":"false",
      "utilityRemittance":"false",
      "otherBankAccount":{
        "beftn":"true",
        "rtgs":"false",
        "npsb":"false"
      },
      "mtbAccount":"true",
      "cash":"true",
      "wallet":{
        "bKash":"true",
        "Nagad":"true",
        "Upay":"true",
        "iPay":"false"
      }
    }
  }
}
```

5.3 Change Password

Controller: extension

End Point: {URL}:<PORT>/<AliasName>/<ControllerName>/changePassword

Authorization: Bearer Token

Method: POST

Request Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M	Request ID. Alphanumeric in format with no special characters. Indicates the unique identifier of the request message generated by Partner.	N	This Id will be used to handling duplicate & end to end message tracing & trouble-shooting. (Every API must have this id and it must be unique for whole life cycle)
newPassword	string	M	Must be at least 16 digit	Y	

Example of Request Body

```
{
  "reqId": "MTB1234567890Aqsrt",
  "newPassword": "Password@1234567" // --- will be encrypted
}
```

Response Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M		N	Request ID received in the corresponding request Body
respCode	string	M		N	RS0000 for success
respDesc	string	M		N	
responseDetails	object				
message	string	O			

Example of Response Body

```
{
  "reqId": "MTB1234567890Aqsrt ",
  "respCode": "RS0000",
  "respDesc": "Request Successful",
  "responseDetails": {
    "message": "Password Changed Successfully"
  }
}
```

5.4 Account Validation

Controller: validate

End Point: {URL}<PORT>/<AliasName>/<ControllerName>/accountValidation

Authorization: Bearer Token

Method: POST

Request Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M	Request ID. Alphanumeric in format with no special characters. Indicates the unique identifier of the request message generated by Partner.	N	This Id will be used to handling duplicate & end to end message tracing & trouble-shooting. (Every API must have this id and it must be unique for whole life cycle)
accountNo	string	M	MTB or Wallet Account Validation	Y	+8801***** for Wallet MSISDN 13-17 digit for MTB Account
beneFirstName	string	M	Beneficiary First Name	N	
beneLastName	string	M	Beneficiary Last Name	N	
beneFullName	string	M	Beneficiary Full Name	N	
accountType	string	M		N	Bank or Wallet
accountOwner	string	M		N	MTB/bKash/Nagad/Upay/iPay

Example of Request Body

MTB

```
{
  "reqId": "MTB1234567890Aqsrt",
  "accountNo": "0460310075297", //--- will be Encrypted
  "beneFirstName": "Lionel",
  "beneLastName": "Messi",
  "beneFullName": "Lionel Messi",
  "accountType": "Bank",
  "accountOwner": "MTB"
}
```

MFS

```
{
  "reqId": "MTB1234567890Aqsrt",
  "accountNo": "+8801717000000", //--- will be Encrypted
  "beneFirstName": "Lionel",
  "beneLastName": "Messi",
  "beneFullName": "Lionel Messi",
  "accountType": "Wallet",
  "accountOwner": "bKash" //--- Nagad/Upay/iPay
}
```

Response Body :

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M		N	Request ID received in the corresponding request Body
respCode	string	M		N	RS0000 for success
respDesc	string	M		N	
responseDetails	object				
accountType	string	M		N	As received in the corresponding request Body
accountOwner	string	M		N	As received in the corresponding request Body
accountNo	string	O		Y	
accountName	string	O		Y	Beneficiary Name received from MTB CBS/bKash/Nagad/Upay/iPay
accountStatus	string	O		N	
dailyTxnCount	string	O		N	
monthlyTxnCount	string	O		N	
lastTxnDateTime	string	O	YYYYMMDDHHMMSS	N	
remarks	string	O		N	

*Example of Response Body***MTB**

```
{
  "reqId": "MTB1234567890Aqsrt",
  "respCode": "RS0000",
  "respDesc": "Request Successful",
  "responseDetails": {
    "accountType": "Bank",
    "accountOwner": "MTB",
    "accountNo": "0460310075297", //----- will be Encrypted
    "accountName": "MD. Osman Habib", //----- will be Encrypted
    "accountStatus": "Operative",
    "dailyTxnCount": "5",
    "monthlyTxnCount": "15",
    "lastTxnDateTime": "20220629105631"
  },
  "remarks": ""
}
```

Wallet

```
{
  "reqId": "MTB1234567890Aqsrt",
  "respCode": "RS0000",
  "respDesc": "Request Successful",
  "responseDetails": {
    "accountType": "Wallet",
    "accountOwner": "bKash", //--- Nagad/Upay/iPay based on Owner
  }
}
```

```

    "accountNo": "01717000000", //---- will be Encrypted
    "accountName": "Jamal Bhuiya", //---- will be Encrypted
    "accountStatus": "Operative",
    "dailyTxnCount": "5",
    "monthlyTxnCount": "15",
    "lastTxnDateTime": "20220629105631",
    "remarks": "VL0Avb2ARM9dCuhn41"
  }
}

```

5.5 Bill Information Validation

Controller: validate

End Point: {URL}<PORT>/<AliasName>/<ControllerName>/billValidation

Authorization: Bearer Token

Method: POST

Request Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M	Request ID. Alphanumeric in format with no special characters. Indicates the unique identifier of the request message generated by Partner.	N	This Id will be used to handling duplicate & end to end message tracing & trouble-shooting. (Every API must have this id and it must be unique for whole life cycle)
billerId	string	M	Bill Number or Student Id or Customer Number or Client Id etc	Y	
billType	string	O	Bill Payment Type	N	Registration/Admission/Enrollment/Without Tax
billingCycle	string	O	Billing Year and Billing Month	N	YYYYMM
billAmount	string	M	Bill or Collection Amount	Y	
userMSISDN	string	M	Account Holder's Mobile Number	Y	
billOwner	string	M	The entity which will receive the Bill (Case Sensitive)	N	DESCO/DPDC-POST/ DPDC-PRE/WASA
billingParams	object	O	Indicates the dynamic parameters. Multiple Data as key, value format.	N	This field is used for splicing services that require special fields.

Example of Request Body

```

{
  "reqId": "MTB1234567890Aqsrt",
  "billerId": "123456789", //--- will be Encrypted
  "billType": "Registration",
  "billingCycle": "202402",
  "billAmount": "500", //--- will be Encrypted
  "userMSISDN": "01919010101",
  "billOwner": "DESCO-POST",
  "billingParams": {}
}

```

Response Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M		N	Request ID received in the corresponding request Body
respCode	string	M		N	RS0000 for success
respDesc	string	M		N	
responseDetails	object				
conversationId	string	M	This Id will be required for Payment Request API	Y	Will expire after 5 minutes
billerId	string	M		Y	As received in the corresponding request Body
billOwner	string	M		N	As received in the corresponding request Body
billingCycle	string	M	Billing Year and Billing Month	N	YYYYMM
customerName	string	M		Y	
departmentId	string	C	Zone Id/ Semester Id		Mandatory for DESCO/DPDC or NSU
billType	string	M	Revenue/Security Deposit/Miscellaneous/Materials	N	Retrieved from Partner API. Returns same value as like in the corresponding request Body if Partner API returns no value
billAmountDetails	object				
billAmount	string	M		Y	Original Bill Amount
meterCharge	string	C		Y	Required for specific Partner
sourceTax	string	C		Y	Required for specific Partner
surChargeAmount	string	C		Y	Required for specific Partner
vatAmount	string	C		Y	Required for specific Partner
tariffAmount	string	C		Y	Required for specific Partner
totalAmount	string	M	Sum of all Amount fields	Y	Actual Bill to be paid
dueDate	string	O		N	YYYYMMDD
issueDate	string	O		N	YYYYMMDD
locationCode	string	O		N	Mandatory for DPDC

*Example of Response Body***MTB**

```

{
  "reqId":"MTB1234567890Aqsr",
  "respCode":"RS0000",
  "respDesc":"Request Successful",
  "responseDetails":{
    "conversationId":"MTB-123456",
    "billerId":"123456789",
    "billOwner":"DESCO-POST",
    "billingCycle":"202402",
    "customerName":"Md. Rajib Bin Ahmed",
    "departmentId":"123456",
    "billType":" Miscellaneous",
    "billAmountDetails":{
      "billAmount":"1200.00",
      "meterCharge":" 0.00",

```

//----- will be Encrypted
//----- will be Encrypted
//----- will be Encrypted
//----- will be Encrypted
//----- will be Encrypted


```

    "sourceTax":"0.00",
    "surChargeAmount":"0.00",
    "vatAmount":"0.00",
    "tariffAmount":"0.00",
    "totalAmount":"1200.00"
  },
  "dueDate":"20240222",
  "issueDate":"20240204",
  "locationCode":"F5"
}
}

```

//---- will be Encrypted
 //---- will be Encrypted
 //---- will be Encrypted
 //---- will be Encrypted
 //---- will be Encrypted

5.6 Payment Request

Controller: Payment

End Point: {URL}<PORT>/<AliasName>/<ControllerName>/PaymentRequest

Authorization: Bearer Token

Method: POST

Request Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M	Request ID. Alphanumeric in format with no special characters. Indicates the unique identifier of the request message generated by Partner.	N	This Id will be used to handling duplicate & end to end message tracing & trouble-shooting. (Every API must have this id and it must be unique for whole life cycle)
referenceNumber	string	M	Pin Number (15 Digit)/ Transaction Id	Y	Unique for Partner's whole lifecycle
tranMode	string	M	WALLET, MTB, OTHERBANK, CASH, UTILITY	N	Based on the value, validations and other criteria will be checked. It's important for Partner to provide transaction Mode correctly.
paymentInfo	object				
remitType	string	M	Default value will be provided	N	WE01- Wage Earner Remittance SR02-Service Export Remittance UR03-Utility Remittance GR04 –Goods Export Remittance
tranAmount	string	M		Y	2 decimal point
originatingCurrency	string	M		N	Sending Currency, Length 3
forexRate	string	M		N	
originatingAmount	string	M		N	
sourceOfFund	string	M		N	
purposeOfFund	string	M		N	
tranRemarks	string	O		N	
senderInfo	object				
senderName	string	M		N	
senderPhone	string	M		N	Must start with + followed by country code
senderGender	string	M	M/F/N	N	M = Male; F=Female ; N = Other
senderOccupation	string	M		N	Service/Business/Others
senderNationality	string	M		N	ISO alpha-2 format

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senderAddressDtIs	object				
sendingCountry	string	M		N	ISO alpha-2 format
senderDistrict	string	O		N	
senderAddress	string	M		N	
beneficiaryInfo	object				
beneficiaryName	string	M		N	
beneficiaryPhone	string	M		N	+8801717000000 Must start with + followed by country code
beneficiaryGender	string	M	M/F/N	N	M = Male; F=Female ; N = Other
beneficiaryOccupati on	string	M		N	Service/Business/Others
beneficiaryAddress	string	M		N	
beneRelationship	string	M	Relationship with Beneficiary	N	
mtbAccount	object		When tranMode = MTB, this object need to have value		
beneficiaryAccount	string	M	Beneficiary MTB Account Number	Y	13-17 Digit
otherBank	object		When tranMode = OTHERBANK, this object need to have value		
tranChannel	string	M	BEFTN/NPSB/RTGS	N	
beneficiaryAccount	string	M	Beneficiary A/C Number in Other Bank	Y	Max 17 digit
bankName	string	O	Beneficiary Bank Name	Y	
branchName	string	O	Beneficiary Bank's Branch Name	Y	
routingNumber	string	M		Y	
walletAccount	object		When tranMode = WALLET, this object need to have value		
beneficiaryAccount	string	M	Beneficiary Wallet Number	Y	+8801717000000 Must start with + followed by country code
accountOwner	string	M	Name of the Wallet Owner	N	bKash/Nagad/Upay/iPay
senderInfoExtended	object		When accountOwner = bKash, this object need to have value		
sendingCorridor	string	O		N	
senderFirstName	string	M		N	
senderLastName	string	M		N	
senderDob	string	O	Sender's Date of Birth	N	YYYYMMDD
senderPob	string	O	Sender's Place of Birth	N	
documentType	string	O		N	
documentNumber	string	O		N	
idIssueDate	string	O	YYYYMMDD	N	
idExpiryDate	string	O	YYYYMMDD	N	
paymentInstrumen tInfo	object		Mandatory for bKash		
piType	string	O		N	
piEntity	string	M		N	
piNumber	string	O		N	
piCity	string	O		N	
piZipCode	string	O		N	
piMisscellExpiry	string	O		N	
piPayonDate	string	O	YYYYMMDD	N	
piMessage	string	O		N	
utilityCollection	object		When tranMode = UTILITY, this object need to have value		
billOwner	string	M	The entity which will receive the Bill (Case Sensitive)	N	DESCO/DPDC-POST/ DPDC-PRE/WASA
billConversationId	string	M	Received as a response param in Bill Information API for specific billOwner	Y	This Id expires after 5 Minutes.

Example of Request Body

MTB Account Transaction

```
{
  "reqId": "MTB1234567890Aqsrt",
  "referenceNumber": "R1234567890",
  "tranMode": "MTB",
  "paymentInfo": {
    "remitType": "WE01",
    "tranAmount": "125.25",
    "originatingCurrency": "EUR",
    "forexRate": "100",
    "originatingAmount": "1.25",
    "sourceOfFund": "Salary",
    "purposeOfFund": "Education",
    "tranRemarks": ""
  },
  "senderInfo": {
    "senderName": "Lionel Messi",
    "senderPhone": "+91234567890",
    "senderGender": "M",
    "senderOccupation": "Service",
    "senderNationality": "AR",
    "senderAddressDtls": {
      "sendingCountry": "FR",
      "senderDistrict": "Paris",
      "senderAddress": "Eiffel Tower"
    }
  },
  "beneficiaryInfo": {
    "beneficiaryName": "Jamal Bhuiya",
    "beneficiaryPhone": "+8801712345678",
    "beneficiaryGender": "M",
    "beneficiaryOccupation": "Service",
    "beneficiaryAddress": "111, MTB Tower",
    "beneRelationship": "Mentor"
  },
  "mtbAccount": {
    "beneficiaryAccount": "1311000773294"
  }
}
```

For BEFTN Transaction

```
{
  "reqId": "MTB1234567890Aqsrt",
  "referenceNumber": "R1234567890",
  "tranMode": "BEFTN",
  "paymentInfo": {
    "remitType": "WE01",
    "tranAmount": "125.25",
    "originatingCurrency": "EUR",
    "forexRate": "100",
    "originatingAmount": "1.25",
    "sourceOfFund": "Salary",

```

```

    "purposeOfFund": "Education",
    "tranRemarks": ""
  },
  "senderInfo": {
    "senderName": "Lionel Messi",
    "senderPhone": "+91234567890",
    "senderGender": "F",
    "senderOccupation": "Service",
    "senderNationality": "AR",
    "senderAddressDtls": {
      "sendingCountry": "FR",
      "senderDistrict": "Paris",
      "senderAddress": "Eiffel Tower"
    }
  },
  "beneficiaryInfo": {
    "beneficiaryName": "Jamal Bhuiya",
    "beneficiaryPhone": "+8801712345678",
    "beneficiaryGender": "F",
    "beneficiaryOccupation": "Service",
    "beneficiaryAddress": "111, MTB Tower",
    "beneRelationship": "Mentor"
  },
  "otherBank": {
    "tranChannel": "BEFTN",
    "beneficiaryAccount": "1311000773294",
    "bankName": "Brac Bank Limited",
    "branchName": "Gulshan",
    "routingNumber": "123456789"
  }
}

```

For Cash Transaction

```

{
  "reqId": "MTB1234567890Aqsrt",
  "referenceNumber": "R1234567890",
  "tranMode": "CASH",
  "paymentInfo": {
    "remitType": "WE01",
    "tranAmount": "125.25",
    "originatingCurrency": "EUR",
    "forexRate": "100",
    "originatingAmount": "1.25",
    "sourceOfFund": "Salary",
    "purposeOfFund": "Education",
    "tranRemarks": ""
  },
  "senderInfo": {
    "senderName": "Lionel Messi",
    "senderPhone": "+91234567890",

```

```

    "senderGender": "N",
    "senderOccupation": "Service",
    "senderNationality": "AR",
    "senderAddressDtls": {
      "sendingCountry": "FR",
      "senderDistrict": "Paris",
      "senderAddress": "Eiffel Tower"
    }
  },
  "beneficiaryInfo": {
    "beneficiaryName": "Jamal Bhuiya",
    "beneficiaryPhone": "+8801712345678",
    "beneficiaryGender": "N",
    "beneficiaryOccupation": "Service",
    "beneficiaryAddress": "111, MTB Tower",
    "beneRelationship": "Mentor"
  }
}

```

For Wallet Transaction

```

{
  "reqId": "MTB1234567890Aqsrt",
  "referenceNumber": "R1234567890",
  "tranMode": "WALLET",
  "paymentInfo": {
    "remitType": "WE01",
    "tranAmount": "125.25",
    "originatingCurrency": "EUR",
    "forexRate": "100",
    "originatingAmount": "1.25",
    "sourceOfFund": "Salary",
    "purposeOfFund": "Education",
    "tranRemarks": ""
  },
  "senderInfo": {
    "senderName": "Lionel Messi",
    "senderPhone": "+91234567890",
    "senderGender": "M",
    "senderOccupation": "Service",
    "senderNationality": "AR",
    "senderAddressDtls": {
      "sendingCountry": "FR",
      "senderDistrict": "Paris",
      "senderAddress": "Eiffel Tower"
    }
  },
  "beneficiaryInfo": {
    "beneficiaryName": "Jamal Bhuiya",
    "beneficiaryPhone": "+8801712345678",
    "beneficiaryGender": "M",
    "senderOccupation": "Service",

```

```

    "beneficiaryAddress": "111, MTB Tower",
    "beneRelationship": "Mentor"
  },
  "walletAccount": {
    "beneficiaryAccount": "+8801712345678",
    "accountOwner": "bKash",           // Upay/Nagad Based on Owner
    "senderInfoExtended": {           // Only required for bKash
      "sendingCorridor": "EUR",
      "senderFirstName": "Lionel",
      "senderLastName": "Messi",
      "senderDob": "19870624",
      "senderPob": "Buenos Aires",
      "documentType": "I",
      "documentNumber": "26105002615",
      "idIssueDate": "",
      "idExpiryDate": ""
    },
    "paymentInstrumentInfo": {         // Only required for bKash
      "piType": "",
      "piEntity": "MoneyGram",
      "piNumber": "",
      "piCity": "",
      "piZipCode": "",
      "piMisscellExpiry": "",
      "piPayonDate": "",
      "piMessage": ""
    }
  }
}

```

Utility Collection Transaction

```

{
  "reqId": "MTB1234567890Aqsrt",
  "referenceNumber": "R1234567890",
  "tranMode": "UTILITY",
  "paymentInfo": {
    "remitType": "UR03",
    "tranAmount": "125.25",
    "originatingCurrency": "EUR",
    "forexRate": "100",
    "originatingAmount": "1.25",
    "sourceOfFund": "Salary",
    "purposeOfFund": "Utility",
    "tranRemarks": ""
  },
  "senderInfo": {
    "senderName": "Lionel Messi",
    "senderPhone": "+91234567890",
    "senderGender": "M",
    "senderOccupation": "Service",
    "senderNationality": "AR",

```

```

    "senderAddressDtls":{
      "sendingCountry":"FR",
      "senderDistrict":"Paris",
      "senderAddress":"Eiffel Tower"
    }
  },
  "beneficiaryInfo":{
    "beneficiaryName":"Jamal Bhuiya",
    "beneficiaryPhone":"+8801712345678",
    "beneficiaryGender":"M",
    "beneficiaryOccupation":"Service",
    "beneficiaryAddress":"111, MTB Tower",
    "beneRelationship":"Mentor"
  },
  "utilityCollection":{
    "billOwner":"DESCO-PRE"
    "billConversationId":"MTB-1213456"
  }
}

```

Response Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M		N	Request ID received in the corresponding request Body
respCode	string	M		N	RS0000 for success
respDesc	string	M		N	
responseDetails	object				
pinNumber	string	M		Y	
paymentMode	string	M		N	WALLET, MTB, OTHERBANK, CASH, UTILITY
paymentOwner	string	M		N	BEFTN/NPSB/RTGS or bKash/Nagad/Upay or Personal or Beneficiary or DESCO/DPDC/WASA/TOPIUP
paymentType	string	M		N	REGULAR/RETURN
paymentState	string	M		N	ACCEPTED/FAILED
paymentStateDesc	string	M		N	

Example of Response Body

```

{
  "reqId":"MTB1234567890Aqsrt",
  "respCode":"RS0000",
  "respDesc":"Request Successful",
  "responseDetails":{
    "pinNumber":"R123456789",
    "paymentMode":"MTB",
    "paymentOwner":"PERSONAL",
    "paymentType":"REGULAR",
    "paymentState":"ACCEPTED",
    "paymentStateDesc":"Payment Request Received"
  }
}

```

5.7 Status Query

Controller: Payment

End Point: {URL}<PORT>/<AliasName>/<ControllerName>/StatusQuery

Authorization: Bearer Token

Method: POST

Request Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M	Request ID. Alphanumeric in format with no special characters. Indicates the unique identifier of the request message generated by Partner.	N	This Id will be used to handling duplicate & end to end message tracing & trouble-shooting. (Every API must have this id and it must be unique for whole life cycle)
page	string	M	User needs to provide Page Number	N	Every request will contain max 1000 transactions. To get the next list of transactions User needs to call this API again with incremented page number.
referenceNumber	string	O	Case Sensitive	Y	
fromDate	string	O	YYYYMMDD	N	
toDate	string	O	YYYYMMDD	N	
paymentMode	string	O	Case Sensitive	N	WALLET, MTB, OTHERBANK, CASH, UTILITY
paymentOwner	string	O		N	BEFTN/NPSB/RTGS or bKash/Nagad/Upay/iPay or Personal or Beneficiary or DESCO/DPDC/WASA/TOUP
paymentState	string	O	Case Sensitive	N	ACCEPTED/LOCKED/EXECUTED/COMPLETED/FAILED/RETURNED/REVERSED/NOTFOUND
sortBy	string	O	Sort by Payment Date; Case Sensitive		ASC/DESC

Example of Request Body

```
{
  "reqId": "MTB0117",
  "page": "1",
  "referenceNumber": "",
  "fromDate": "",
  "toDate": "",
  "paymentMode": "",
  "paymentOwner": "",
  "paymentState": "",
  "sortBy": "ASC"
}
```

Response Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M		N	Request ID received in the corresponding request Body

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respCode	string	M		N	RS0000 for success
respDesc	string	M		N	
responseDetails	object				
contentDetails	object				
currentPage	string	M		Y	As received in the corresponding request
totalPages	string	M			Number of pages to show for the transactions found
pageSize	string	M			1000 transactions history in a page
trxCount	string	M			Number of transactions showing in this particular response
totalTrxCount	string	M			Total number of transactions for the given request
transactionDetails	list				
pinNumber	string	M		Y	
paymentAmount	string	M		Y	
paymentMode	string	M		N	WALLET, MTB, OTHERBANK, CASH, UTILITY
paymentOwner	string	M		N	BEFTN/NPSB/RTGS or bKash/Nagad/Upay/iPay or Personal or Beneficiary or DESCO/DPDC/WASA/TOPIUP
paymentType	string	M		N	REGULAR/RETURN
paymentState	string	M		N	ACCEPTED/LOCKED/EXECUTED/COMPLETED /FAILED/CANCELED/RETURNED/REVERSED
paymentStateDesc	string	M		N	
paymentDate	string	M	YYYYMMDDHH24MISS	N	

Example of Response Body

```
{
  "reqId": "MTB0117",
  "respCode": "RS0000",
  "respDesc": "Request Successful",
  "responseDetails": {
    "ContentDetails": {
      "currentPage": "1",
      "totalPages": "1",
      "pageSize": "1000",
      "trxCount": "23",
      "totalTrxCount": "23"
    },
    "TransactionDetails": [
      {
        "pinNumber": "R00000001",
        "paymentAmount": "100.65",
        "paymentMode": "MTB",
        "paymentOwner": "PERSONAL",
        "paymentType": "REGULAR",
        "paymentState": "ACCEPTED",
        "paymentStateDesc": "Payment Request Received",
        "paymentDate": "20220731115750"
      },
      {
        "pinNumber": "R00000002",
        "paymentAmount": "100.65",

```

```

    "paymentMode": "OTHERBANK",
    "paymentOwner": "BEFTN",
    "paymentType": "REGULAR",
    "paymentState": "ACCEPTED",
    "paymentStateDesc": "Payment Request Received",
    "paymentDate": "20220731121048"
  },
  {
    "pinNumber": "R00000003",
    "paymentAmount": "100.65",
    "paymentMode": "OTHERBANK",
    "paymentOwner": "NPSB",
    "paymentType": "REGULAR",
    "paymentState": "ACCEPTED",
    "paymentStateDesc": "Payment Request Received",
    "paymentDate": "20220731121048"
  },
  {
    "pinNumber": "R00000005",
    "paymentAmount": "100.00",
    "paymentMode": "CASH",
    "paymentOwner": "BENEFICIARY",
    "paymentType": "REGULAR",
    "paymentState": "ACCEPTED",
    "paymentStateDesc": "Payment Request Received",
    "paymentDate": "20220731133257"
  },
  {
    "pinNumber": "R00000006",
    "paymentAmount": "100",
    "paymentMode": "WALLET",
    "paymentOwner": "bKash",
    "paymentType": "REGULAR",
    "paymentState": "ACCEPTED",
    "paymentStateDesc": "Payment Request Received",
    "paymentDate": "20220731133348"
  },
  {
    "pinNumber": "R00000100",
    "paymentAmount": "1200.00",
    "paymentMode": "UTILITY",
    "paymentOwner": "DESCO-PRE",
    "paymentType": "REGULAR",
    "paymentState": "ACCEPTED",
    "paymentStateDesc": "Payment Request Received",
    "paymentDate": "20220731115750"
  }
]
}
}

```

5.8 Account Statement

Controller: Extension

End Point: {URL}:<PORT>/<AliasName>/<ControllerName>/AccountStatement

Authorization: Bearer Token

Method: POST

Request Body:

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M	Request ID. Alphanumeric in format with no special characters. Indicates the unique identifier of the request message generated by Partner.	N	This Id will be used to handling duplicate & end to end message tracing & trouble-shooting. (Every API must have this id and it must be unique for whole life cycle)
accountNumber	string	M	13 Digit Account Number	Y	0460310001234
fromDate	string	M	YYYYMMDD	N	20231231
toDate	string	M	YYYYMMDD	N	20231231
sortBy	string	M	Sort by Payment Date; Case Sensitive	N	ASC/DESC

Example of Request Body

```
{
  "reqId": "MTB0138",
  "accountNumber": "0460310001234",
  "fromDate": "20231001",
  "toDate": "20231001",
  "sortBy": "ASC"
}
```

Response Body :

Parameter	Data Type	Mandatory /Optional	Description	Encryption	Remarks
reqId	string	M		N	Request ID received in the corresponding request Body
respCode	string	M		N	RS0000 for success
respDesc	string	M		N	
responseDetails	object				
transactionDetails	list				
transactionDate	string	M	Tran Date in Core Banking System	N	DD-MM-YYYY HH24:MI
withdraw	string	M	Tran Amount if transactionType = Debit	N	
deposit	string	M	Tran Amount if transactionType =Credit	N	
transactionType	string	M	Defines the type of Accounting	N	Debit/Credit
narration	string	M	Transaction Remarks. String before First pipe () will be the Transaction Reference Number	N	690930907 R BKASH +8801660163971 SAM SUN NAHAR
currentBalance	string	M	Available Balance after the particular Transaction happened.	N	

Example of Response Body

```
{
  "reqId": "MTB0143",
  "respCode": "RS0000",
  "respDesc": "Request Successful",
  "responseDetails": {
    "transactionList": [
      {
        "transactionDate": "01-10-2023 03:17",
        "withdraw": "496.43",
        "deposit": "0.00",
        "transactionType": "Debit",
        "narration": "690930907|R|BKASH|+8801660163971|SAMSUN NAHAR",
        "currentBalance": "180122766.03"
      },
      {
        "transactionDate": "01-10-2023 03:21",
        "withdraw": "3000.00",
        "deposit": "0.00",
        "transactionType": "Debit",
        "narration": "691018777|R|BKASH|+8801646310699|MST SAYLA KHANO",
        "currentBalance": "179192779.93"
      },
      {
        "transactionDate": "01-10-2023 03:22",
        "withdraw": "0.00",
        "deposit": "7000.00",
        "transactionType": "Credit",
        "narration": "691020626|REV|BKASH|+8801984562264|MST MONZIRA RAH",
        "currentBalance": "179185779.93"
      },
      {
        "transactionDate": "01-10-2023 03:22",
        "withdraw": "2089.24",
        "deposit": "0.00",
        "transactionType": "Debit",
        "narration": "691029321|R|BKASH|+8801725573788|MD RAJU AHAMED",
        "currentBalance": "179163791.8"
      },
      {
        "transactionDate": "01-10-2023 13:22",
        "withdraw": "33278.00",
        "deposit": "0.00",
        "transactionType": "Debit",
        "narration": "691032658|R|BKASH|+8801610346759|ARSHEE AFRIDA K",
        "currentBalance": "179130513.8"
      }
    ]
  }
}
```

Appendix

6.1 Document Control

Version	Date	Author(s)	Comments
0.1	01.01.2023	Tamim B.	Initial Draft
0.2	01.03.2023	Tamim B.	Peer Review
1.0	01.05.2023	Tamim B.	1 st Release
1.1	25.05.2023	Tamim B.	Payment Request API modification
2.0	21.06.2023	Tamim B.	Document Update
2.1	24.08.2023	Tamim B.	Sender Country and Nationality are set to ISO 2 format. Document Update.
2.2	28.12.2023	Tamim B.	<u>Account Statement</u> API added
3.0	11.02.2024	Tamim B.	- New Wallet Partner Added, Named as UPAY - New Remittance Type Added as Utility Remittance (UR03) - New Parameter Added in Exchange House Details API Response (tranModeActive -> utilityRemittance && tranModeActive.wallet -> Upay) - New Example added in <u>Account Validation</u> API Response for Upay Wallet - New API Added as <u>Utility Bill Information</u>. - New Parameter Object Added in <u>Payment Request</u> API for Utility Bill Payment. - Request and Response Parameter Example added for Utility Bill Payment in <u>Payment Request</u> API. - New Parameter Added in <u>Status Query</u> API Request as paymentOwner. - New Parameter Added in <u>Status Query</u> API Response as paymentAmount & paymentOwner. - Request and Response Parameter Example modified for <u>Status Query</u> API.